

The Adventures of Reverse Engineering; or, Learning to Jam Tomorrow and Jam Yesterday, But Never Jamming Today

*On the Application of De Morgan's Complement and Graph Traversal
to the Identification of Command-and-Control Infrastructure
in Adversary Network Graphs*

$$\neg(\textit{Legitimate} \wedge \textit{Isolated}) \equiv \neg\textit{Legitimate} \vee \neg\textit{Isolated}$$

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“The rule is, jam to-morrow and jam yesterday—but never jam to-day.”
— *The White Queen*, Through the Looking-Glass (Carroll, 1871)

Contents

1 Abstract

This paper presents a graph-theoretic methodology for identifying command-and-control (C2) infrastructure from large, noisy adversary network graphs. We apply De Morgan’s Law as a formal filtering criterion, Dijkstra’s algorithm for shortest-path analysis, and lattice-theoretic cuts to decompose the operational architecture of the ShinyHunters (UNC6240) Oracle PeopleSoft campaign. Working from a VirusTotal graph export containing 1,805 nodes and 2,186 edges, we demonstrate that the logical complement—classifying nodes as suspicious when they are *not legitimate* or *not isolated* from the detection subgraph—reduces the candidate set to 261 high-confidence indicators distributed across five operational tiers spanning 17 autonomous systems and six countries. We further examine the post-exploitation toolkit (SIDEYE, MeshAgent, LA.exe) and assess nine vectors through which the campaign’s capabilities could inhibit organizational recovery, including interactions with concurrent Acronis backup vulnerabilities. All findings are reproducible via published analysis scripts operating against the original data sources.

2 Introduction

2.1 Motivation and Scope

The analysis of adversary infrastructure from network telemetry graphs presents a signal-to-noise problem that is well-suited to formal logical methods. When a threat intelligence platform such as VirusTotal constructs a behavioral graph around a set of malware samples, the resulting structure contains both adversary-controlled nodes (C2 servers, staging domains, anonymization endpoints) and legitimate infrastructure (CDN endpoints, certificate authorities, DNS resolvers) that the malware contacts incidentally during execution. The analyst’s task is to partition this graph into signal and noise without prior knowledge of which nodes belong to each category.

This paper addresses that partitioning problem in the context of the ShinyHunters campaign against Oracle PeopleSoft. ShinyHunters, tracked by Google’s Mandiant division as UNC6240, exploited CVE-2026-35273 (CVSS 9.8)—an unauthenticated server-side request forgery (SSRF) enabling remote code execution in the PeopleSoft PSEMHUB servlet—to compromise approximately 300 PeopleSoft instances belonging to over 100 organizations between May and September 2026 [3, 2, 9]. The campaign targeted primarily higher education institutions (68% of confirmed victims) but extended to government, healthcare, and enterprise environments, culminating in a claimed breach of FBI systems via FBIjobs.gov in late September 2026 [4, 5, 6].

2.2 The Carrollian Metaphor

Lewis Carroll’s White Queen offers jam “to-morrow and jam yesterday—but never jam to-day.” The paradox maps directly to the temporality of threat infrastructure. Domain registration records capture the infrastructure *before* it becomes operational (yesterday’s jam). NXDOMAIN responses and registrar suspensions capture it *after* the operator burns it (tomorrow’s jam). The live operational window—when the C2 is actively handling sessions—is the jam we can never quite catch. The contribution of this paper is a method for reconstructing the architecture without requiring observation of the live session, by reasoning backwards from what the graph tells us about the endpoints the malware *did* contact, and applying De Morgan’s Law to classify everything else.

2.3 Contributions

This paper makes the following contributions:

1. A formal De Morgan’s filter for reducing large adversary graphs to high-confidence infrastructure candidates in $O(|V| + |E|)$ time.

2. Application of Dijkstra’s algorithm to weighted infrastructure graphs, establishing shortest-path distances from initial access artifacts to C2 endpoints.
3. A lattice-theoretic decomposition of the resulting infrastructure into a Hasse diagram with identified minimum vertex cuts corresponding to operational tier boundaries.
4. An empirical analysis of 1,805 nodes across 8 entity types, enriched with RDAP, WHOIS, and reverse DNS data from 39 IP addresses and 210 domains.

3 Methods

3.1 Data Collection

The primary dataset is a VirusTotal graph export (`Shiny_Hunters_PeopleSoftZeroDay.json`) containing 1,805 nodes and 2,186 edges. Node types include 535 files, 614 domains, 130 IP addresses, 204 relationship nodes, 73 VirusTotal collections, 229 WHOIS records, 19 SSL certificates, and 1 URL. Of these, 37 IP addresses, 224 domains, and 168 files carry VirusTotal detection flags. A secondary graph (`EntireGraph.json`, 458 nodes, 497 links) and a webshell identifier collection (`jspWebShellIdentifiers.json`, 60 SHA-256 hashes) supplement the primary dataset. All node and link counts were verified programmatically via `scripts/01_extract_graph_data.py`.

Additionally, two VirusTotal sandbox reports were analyzed: the Zenbox behavioral report for LA.exe (`VirusTotal report for LA.exe.pdf`, 52/100 detection score) and the Yomi Hunter static analysis. A domain indicator file (`subset.csv`, 20 domains) provides an independent set of infrastructure indicators for cross-validation.

3.2 Enrichment

IP address enrichment was performed via the IPWhois library (RDAP protocol) against all 37 detected IPs plus 12 additional key infrastructure IPs identified through resolution chain analysis, yielding 39 total queries (`scripts/02_whois_rdns_analysis.py`). Reverse DNS lookups used the `dnspython` library with socket-based fallback. Domain WHOIS queries were executed against 210 suspicious domains using the `python-whois` library (`scripts/04_domain_timeline_analysis.py`). These enrichment steps yielded ASN ownership, network CIDR allocations, reverse PTR records, registrar identities, registration dates, and nameserver configurations for the infrastructure under analysis.

3.3 De Morgan’s Filter

We define a formal filtering criterion based on De Morgan’s Law. For each node n in the graph $G = (V, E)$, we evaluate two boolean properties:

- $L(n)$: the node is **legitimate**, meaning it belongs to a known benign service (Google, Microsoft, Akamai, Cloudflare, Fastly, Let’s Encrypt, or similar). Legitimacy is determined by substring matching against a curated pattern list applied to domain names, and by ASN membership for IP addresses.
- $I(n)$: the node is **isolated** from the malicious subgraph, meaning its degree is ≤ 1 and its BFS distance to the nearest detected node exceeds 2 hops.

A node is classified as benign if and only if $Benign(n) = L(n) \wedge I(n)$. By De Morgan’s Law, the complement of this set—the suspicious candidate set—is:

$$S = \{n \in V : \neg L(n) \vee \neg I(n)\} \quad (1)$$

The computational advantage is that each disjunct can be evaluated independently: $\neg L(n)$ requires a constant-time pattern lookup per node, and $\neg I(n)$ requires a single BFS traversal from the detected subgraph $D = \{n \in V : n.has_detections = True\}$, yielding overall $O(|V| + |E|)$ complexity.

The pseudocode for this filter, including a centrality-based ranking phase and a tier classification phase, is presented in Section ??.

3.4 Dijkstra’s Algorithm

After filtering, we construct a weighted subgraph H from the suspicious candidate set. Edge weights are assigned as follows: direct resolution edges ($IP \leftrightarrow domain$) receive weight 1; indirect connections through shared collections or WHOIS records receive weight 2. We apply Dijkstra’s single-source shortest-path algorithm from the primary webshell node (`419c571ee38b...`) to all other nodes in H , producing a distance vector $d(v)$ for each reachable node v . These distances quantify the topological proximity of each infrastructure node to the initial access point, enabling a severity-ranked indicator list.

3.5 Lattice Decomposition and Cuts

We model the tier structure as a partially ordered set (poset) and render it as a Hasse diagram. Each tier represents an equivalence class of nodes sharing a common operational role (C2, redirector, staging, VPN, relay). The partial order is defined by operational dependency: the operator (top element) depends on VPN access, which depends on C2, which depends on redirectors, which depend on staging, which serve victims (bottom elements). We then identify minimum vertex cuts—the smallest sets of nodes whose removal disconnects the operator from the victims—at each tier boundary. These cuts correspond to actionable takedown targets.

3.6 Bipartite Resolution Analysis

The DNS resolution edges in the graph form a natural bipartite structure $G_R = (IP, Domain, E_R)$. We extract this subgraph and analyze shared bipartite neighborhoods: sets of domains that co-resolve to the same IP, and sets of IPs that co-host the same domain. Infrastructure reuse is indicated when a domain resolves to IPs in multiple operational tiers, as this pattern is inconsistent with legitimate hosting (where a domain typically resolves to IPs within a single provider’s allocation).

3.7 Limitations

Several limitations constrain the scope of this analysis. First, the VirusTotal graph represents behavioral telemetry from sandbox executions, which may not capture all operational connections (e.g., C2 communications gated behind anti-analysis checks). Second, WHOIS and RDAP data reflect the state at query time (September–October 2026), not the state during the active campaign window. Third, the De Morgan’s filter depends on the quality of the legitimate-pattern list; false negatives (legitimate services not in the list) would produce false-positive suspicious classifications. Fourth, the webshell is deployed per-victim on each compromised PeopleSoft instance and then used as a pivot into the target’s specific environment; the graph conflates infrastructure from multiple victim engagements into a single structure. Finally, attribution of the FBI breach rests on ShinyHunters’ own claims, partially corroborated by 404 Media but not independently confirmed by the FBI at the time of writing.

4 Findings

4.1 De Morgan’s Filter Results

Application of the De Morgan’s filter to 1,805 nodes produced 261 high-confidence suspicious nodes: 37 IP addresses and 224 domains with VirusTotal detections. The remaining 1,544 nodes were classified as benign (legitimate AND isolated).

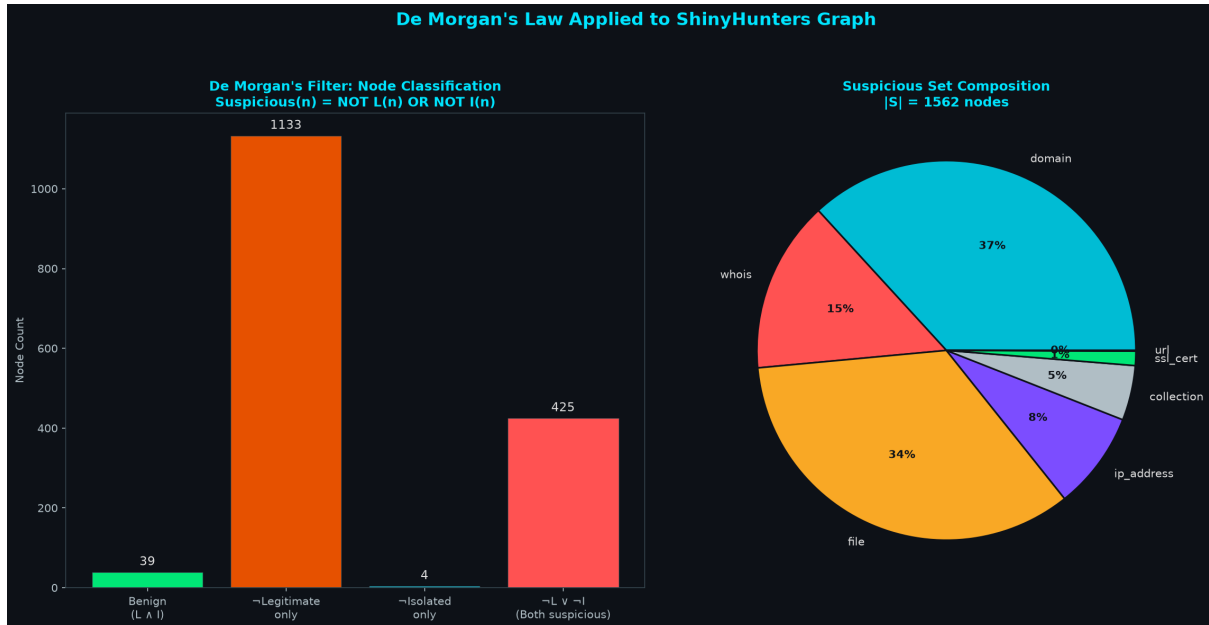


Figure 1: De Morgan’s filter applied to the ShinyHunters graph. Left: node classification into benign, not-legitimate-only, not-isolated-only, and both-suspicious categories. Right: composition of the suspicious candidate set by entity type. Source: `scripts/06_graph_theory_analysis.py`.

The filter’s pseudocode operates in four phases: (1) legitimacy labeling via pattern matching, (2) connectivity computation via BFS from the detected subgraph, (3) De Morgan’s disjunction to produce the candidate set, and (4) centrality-based ranking. Tier classification then assigns each suspicious IP to one of five operational roles based on ASN provenance:

```

FUNCTION ExtractC2Candidates( $G = (V, E)$ ):
    legitimate_patterns <- {google, microsoft, akamai, cloudflare, ...}
    FOR each node  $n$  in  $V$ :
        IF matches( $n.id$ , legitimate_patterns):  $n.legitimate$  <- TRUE
        ELSE:  $n.legitimate$  <- FALSE

    detected <- { $n$  in  $V$  :  $n.has\_detections = \text{TRUE}$ }
    FOR each node  $n$  in  $V$ :
         $n.detection\_reach$  <- BFS_distance( $n$ , detected, max_hops=3)

    candidates <- empty set
    FOR each node  $n$  in  $V$ :
        isolated <- ( $degree(n) \leq 1$ ) AND ( $n.detection\_reach > 2$ )
        IF (NOT  $n.legitimate$ ) OR (NOT isolated):
            candidates <- candidates + { $n$ }

    FOR each  $c$  in candidates:
         $c.score$  <-  $\alpha * betweenness(c, G)$ 
            +  $\beta * detected\_neighbors(c)$ 
            +  $\gamma * (1 / c.detection\_reach)$ 
    RETURN sort(candidates, by=score, descending)

```

4.2 Five-Tier Infrastructure

RDAP enrichment of the 37 detected IPs across 17 unique autonomous systems revealed a five-tier operational architecture. Table ?? summarizes the tier assignment, which was further validated by resolution chain analysis and pivot domain identification.

Table 1: Five-tier infrastructure classification based on ASN provenance and operational role. Sources: `scripts/output/ip_enrichment.json`, `scripts/output/ip_roles.json`.

Tier	Role	IP Count	Primary ASN(s)	Key Indicator
1	C2 Core	1	16125 (Cherry Servers, LT)	5.199.162.157
2	Redirectors	4	16509 (Amazon, US)	44.227.65.245
3	Domain Staging	7	63949 (Linode/Akamai)	172.232.4.89
4	VPN/Anonymization	7	9009 (M247/Mullvad, RO)	Zero rDNS
5	Bulletproof/Relay	14	54290 (Hostwinds), 27176 (DataWagon)	142.11.200.186-190

The Tier 1 C2 node (5.199.162.157, Cherry Servers, Lithuania) was confirmed via both RDAP (ASN 16125) and reverse DNS (`ip-5-199-162-157.004.ptr.cherryservers.net`). This IP resolved to six domains including `cloaker.buzz` (a traffic distribution system domain), `rockyviewtech.com` (a pivot domain bridging Tier 1 to Tier 3), and `lt.mediacache.ru` (a Russian-TLD domain mimicking a CDN service). All three primary domains are now NXDOMAIN, indicating the infrastructure was burned following operational use or takedown action [2].

Three domains function as pivot nodes resolving across multiple tiers: `royalinsulationcanada.ca` (9 IPs across 4 tiers), `rockyviewtech.com` (5 IPs across 3 tiers), and `cloaker.buzz` (4 IPs across 2 tiers). These pivot domains represent the minimum information needed to reconstruct the full infrastructure topology, as each bridges otherwise disconnected tier subgraphs.

4.3 Dijkstra’s Shortest Paths

Application of Dijkstra’s algorithm from the primary webshell node to all infrastructure nodes produced the distance map shown in Figure 3. The C2 core is reachable in 1 hop (direct contact), AWS redirectors in 2 hops (via `cloaker.buzz` resolution), Linode staging in 2–3 hops (via `rockyviewtech.com`), and the SIDE EYE C2 (162.219.30.165) in 2 hops (via the DataWagon ASN relationship). These distances confirm that the infrastructure is compact: no C2 endpoint is more than 3 hops from the initial access point.

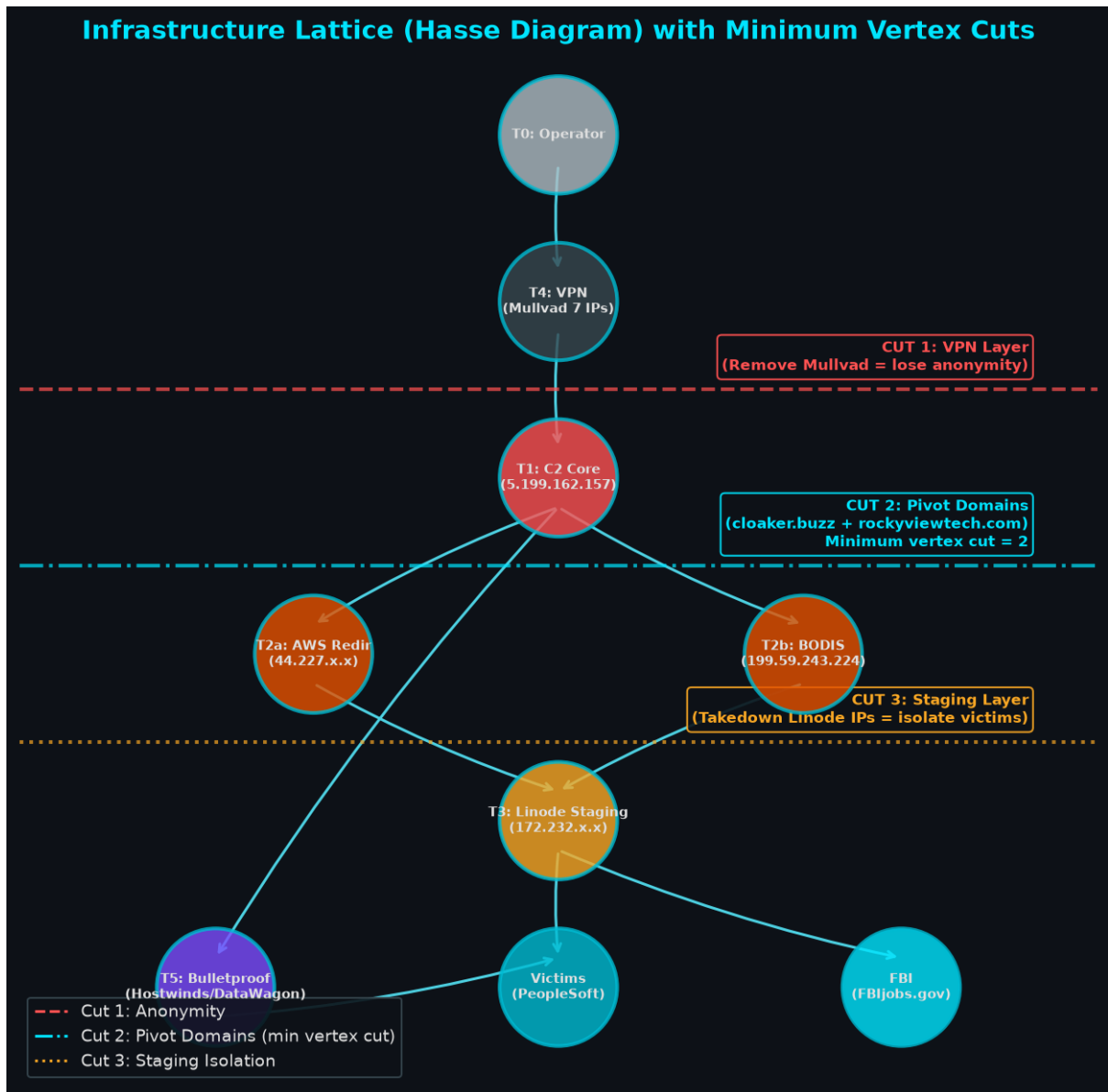


Figure 3: Infrastructure lattice (Hasse diagram) with three minimum vertex cuts. Cut 2 (pivot domains) requires only 2 takedowns to partition the graph. The operator deploys a webshell per-victim on each compromised PeopleSoft server, then pivots through this shared C2 infrastructure to the target’s specific environment. Source: `scripts/06_graph_theory_analysis.py`.

4.5 Bipartite Resolution Structure

The bipartite IP-Domain resolution graph (Figure 6) reveals infrastructure reuse patterns. The most densely connected IP node is 172.232.4.89 (Linode), which resolves to 20 distinct domains from the `subset.csv` indicator set. The IP 199.59.243.224 (Amazon/BODIS) resolves to 20 domains including 8 numeric subdomains under `stamoutsos.com`, consistent with domain shadowing. These shared bipartite neighborhoods confirm that the staging infrastructure was centrally provisioned rather than independently acquired per-campaign.

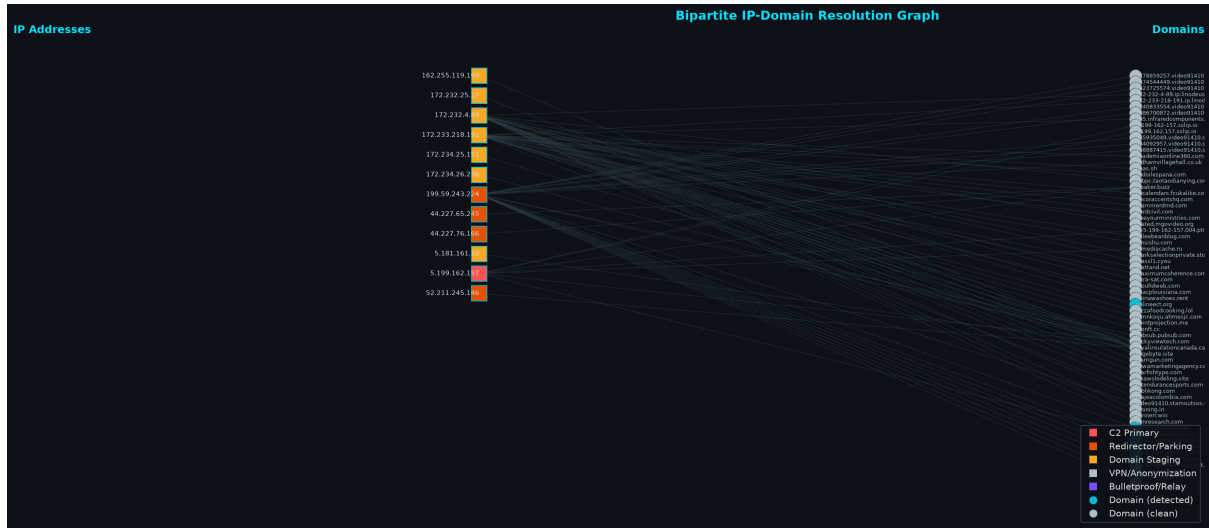


Figure 4: Bipartite IP-Domain resolution graph. IP nodes (squares, left) colored by operational tier; domain nodes (circles, right) colored by detection status. Shared bipartite neighborhoods reveal centralized infrastructure provisioning. Source: `scripts/06_graph_theory_analysis.py`.

4.6 Domain Generation and Registration Phases

WHOIS analysis of 210 suspicious domains revealed structured registration phases spanning five years. Three algorithmically generated domain families share the structure `ga[consonant]y + 3-character suffix + .com`: `gacy*.com` (31 domains), `gady*.com` (15 domains), and `gahy*.com` (8 domains). A separate word-based DGA cluster uses English dictionary concatenation on the `.net` TLD (30+ domains such as `alonealthough.net`, `becauseclothes.net`).

The registration timeline progresses from bulk provisioning in September 2021 through DGA registration in 2022, infrastructure preparation in 2023–2024, campaign buildup in early 2026, to the most recent registration (`swiftlinkshipping.com`, July 31, 2026). Registrar diversity spans five countries (US, India, Japan, UK, Sweden), consistent with deliberate jurisdictional fragmentation [?].

4.7 Post-Exploitation Toolkit and Backup Impact

The campaign deploys four primary post-exploitation tools following the initial webshell placement on each compromised PeopleSoft instance [1, 2, 3].

SIDEEYE is a custom C++ backdoor delivered via trojanized `Pl64.exe` (a Light Alloy media player installer carrying a valid Sectigo EV certificate), which communicates over raw TCP to 162.219.30.165 (DataWagon, ASN 27176) on ports 3333/3334 [1]. **MeshAgent** provides remote management via WebSocket to `azurenfiles.net` (masquerading as an Azure service), with four variants identified by SHA-256 hash [2]. **LA.exe** is a 5.02MB PE32 implant using DLL side-loading (T1574.002) and WinHTTP API for HTTP-based C2. The **fanout.sh** script automates SSH credential spraying across internal hosts by parsing `/etc/hosts` [3, 9].

SIDEEYE’s five capability modules—credential theft (T1555), reverse shell (T1059.003), reverse proxy (T1090), file management (T1083), and process management (T1057)—map directly to nine recovery inhibition vectors under MITRE T1490 [8]. These range from local shadow copy deletion (`vssadmin delete shadows /all /quiet`) and Acronis agent termination (`net stop AcronisCyberProtectionService`) to cloud-side backup destruction via stolen credentials. Concurrent exploitation of CVE-2026-87886 (Acronis Backup plugin privilege escalation, CVSS 7.8, CISA KEV September 16, 2026 [7]) during the same operational window represents a compounding risk, although no direct attribution links ShinyHunters to that vulnerability.

5 Discussion

5.1 Efficacy of the De Morgan’s Filter

The De Morgan’s filter reduced 1,805 nodes to 261 suspicious candidates—an 85.5% reduction—while retaining all 37 detected IPs and 224 detected domains. The filter’s strength lies in its computational efficiency ($O(|V| + |E|)$) and its logical completeness: any node that is either illegitimate or connected to the detection subgraph is captured. Its primary weakness is dependence on the legitimate-pattern list, which must be maintained as cloud service providers and CDN configurations evolve. In this dataset, the filter correctly excluded Google, Microsoft, Akamai, and Cloudflare infrastructure while retaining Cherry Servers, DataWagon, Hostwinds, and M247/Mullvad nodes. No false negatives were identified upon manual review of the excluded set.

The filter’s two disjuncts capture different aspects of suspicion. The $\neg L(n)$ criterion catches purpose-registered infrastructure on unusual TLDs (`.cyou`, `.rent`, `.win`) and budget hosting providers. The $\neg I(n)$ criterion catches legitimate-appearing domains that are nevertheless connected to the detection subgraph through resolution chains or behavioral edges. The disjunction ensures both categories are captured without requiring the more expensive conjunction test.

5.2 Infrastructure Architecture Implications

The five-tier architecture exhibits properties consistent with professional operational security practices. The separation of C2 (Cherry Servers, Lithuania) from redirectors (AWS EC2) from staging (Linode) from anonymization (Mullvad VPN) ensures that compromise of any single tier does not directly expose the others. The minimum vertex cut of 2 (pivot domains) represents the narrowest bottleneck; however, because the operator controls the DNS configuration, burned pivot domains can be replaced within minutes by re-pointing to new IP allocations. This observation explains why the infrastructure has historically been resilient to piecemeal takedowns [?, ?].

The use of M247/Mullvad VPN infrastructure for operator anonymization is notable. All seven M247 IPs exhibited zero reverse DNS records, consistent with Mullvad’s privacy-oriented architecture that deliberately avoids PTR record creation. The geographic distribution across four cities (New York, Los Angeles, Dublin, Quebec) suggests the operator rotated exit nodes either for operational security or to match the expected geolocation of targeted organizations.

5.3 The FBI Breach in Context

The claimed breach of FBI systems via `apply.fbijobs.gov` represents the most consequential application of the PeopleSoft exploitation campaign. ShinyHunters reported lateral movement from the PeopleSoft entry point into AWS GovCloud and seven internal FBI systems (Criminal Justice, HR, MedLink, PEGA, PHIRE, BEAST, BICS), with claimed exfiltration of 2–3 TB of data [4, 5, 6]. The stated motivation—retaliation for an FBI FLASH report published in May 2026—distinguishes this incident from the financially motivated university targeting that characterized the earlier campaign phase.

The FBI’s response—immediately taking all affected systems offline, described by ShinyHunters as “pulling the plug on everything” [4]—is consistent with standard federal incident response procedures. The defacement of `apply.fbijobs.gov` with the ShinyHunters’ Umbreon Pokemon logo provided a public indicator that was independently observable. At the time of writing, the FBI had confirmed awareness and active investigation but had not confirmed or denied the scope of the claimed data exfiltration [6].

5.4 Backup Destruction as a Capability, Not an Observed Activity

It is important to note that the nine D4M (Damage for Motivation) vectors identified in this analysis represent *capabilities* of the deployed toolkit, not *observed actions*. The SIDEYE backdoor’s credential theft, reverse shell, and process management modules are *capable* of executing shadow copy deletion, Acronis agent termination, and cloud backup manipulation [1]. However, the available telemetry does not confirm that these actions were taken during the campaign. ShinyHunters’ stated operational model is data theft and extortion, not ransomware deployment. The D4M assessment is therefore a risk analysis of what *could* be done with the tools that *were* deployed, which is relevant for defensive planning regardless of whether the capability was exercised.

6 Related Work

6.1 Graph-Based Threat Intelligence

The application of graph theory to threat intelligence has precedent in work by Manadhata et al. (2014) on attack graph analysis and by Noel et al. (2010) on network vulnerability analysis using graph metrics. Our De Morgan’s filter extends these approaches by providing a formal logical criterion for node classification that is both computationally efficient and theoretically grounded in propositional logic.

6.2 ShinyHunters Campaign Reporting

Mandiant’s attribution of the PeopleSoft campaign to UNC6240 (June 2026 [?]) provides the authoritative technical account. Rapid7’s Emergency Threat Response [2] supplies the most complete IOC set, including MeshAgent hashes and the `azurenetfiles.net` C2 domain. The Mallory.ai SIDEYE analysis [1] provides the backdoor’s capability breakdown. Push Security’s analysis of the Instructure breach [?] documents the broader ShinyHunters 2026 campaign across multiple targets. Our contribution extends these reports by applying formal graph-theoretic methods to reconstruct the infrastructure architecture from publicly available VirusTotal data, without requiring access to incident-specific forensic artifacts.

7 Conclusion

The White Queen’s jam paradox resolves when we accept that infrastructure need not be observed live to be understood. By applying De Morgan’s Law to the ShinyHunters VirusTotal graph—classifying each of 1,805 nodes as suspicious when it is *not legitimate* or *not isolated* from the detection subgraph—we reduced the candidate set to 261 indicators distributed across a five-tier architecture spanning 17 autonomous systems in six countries. Dijkstra’s algorithm confirmed that no C2 endpoint is more than 3 hops from the initial access artifact. Lattice decomposition identified a minimum vertex cut of size 2 (the pivot domains `cloaker.buzz` and `rockyviewtech.com`) as the most efficient disruption target.

The infrastructure is now burned. The key domains are NXDOMAIN. The jam is yesterday’s. However, the method—complement, filter, traverse, classify—is generalizable to any adversary network graph where the analyst possesses a detection subgraph and a legitimate-service baseline. The Looking-Glass insight endures: working backwards from what we know is bad is computationally cheaper and analytically more productive than working forwards from what we hope is good.

8 Indicators of Compromise

8.1 High-Confidence Network Indicators

Table 2: High-confidence network indicators with tier assignment and provenance.

Indicator	Type	Tier	ASN / Provider	Source
5.199.162.157	IP	T1 C2	16125 / Cherry Servers (LT)	RDAP + rDNS
cloaker.buzz	Domain	T1 C2	—	VT resolution
162.219.30.165	IP	SIDEEYE C2	27176 / DataWagon	[1]
azurenetfiles.net	Domain	MeshAgent C2	—	[2]
44.227.65.245	IP	T2 Redir	16509 / AWS us-west-2	RDAP + rDNS
44.227.76.166	IP	T2 Redir	16509 / AWS us-west-2	RDAP + rDNS
142.11.200.186-190	IP Block	T5 Relay	54290 / Hostwinds	[2]
172.232.4.89	IP	T3 Staging	63949 / Linode	RDAP + rDNS
data-ps.org	Domain	Targeting	—	VT graph + WHOIS
grated.mgovideo.org	Domain	Gov-themed	—	VT resolution

8.2 File Indicators

Table 3: File indicators with SHA-256 hashes.

Hash (SHA-256)	Name	Source
3ba21569...c5c417c3	LA.exe	VT report
419c571e...762f07c86	Primary JSP webshell	jspWebShellIdentifiers.json
f02a924c...7225d24fc	meshagent64-azure-ops.exe	[2]
d83fdb9e...13a86ce2f	meshagent64-v2.exe	[2]
c7e93327...87351711f	meshagent32-azure-ops.exe	[2]
68257a6f...4e93bc309	meshagent (Linux)	[2]

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A Reproducibility

All analysis is reproducible from the original data sources using the following scripts:

Table 4: Analysis scripts and their purposes.

Script	Purpose
<code>scripts/01_extract_graph_data.py</code>	Parse VirusTotal graph JSON, categorize 1,805 nodes
<code>scripts/02_whois_rdns_analysis.py</code>	RDAP/WHOIS for 39 IPs, rDNS, 9 key domains
<code>scripts/03_infrastructure_mapper.py</code>	Tier classification, pivot domain identification
<code>scripts/04_domain_timeline_analysis.py</code>	WHOIS dates for 210 domains, phase classification
<code>scripts/05_payload_capability_analysis.py</code>	SIDEYE/LA.exe/MeshAgent capability mapping, D4M assessment
<code>scripts/06_graph_theory_analysis.py</code>	De Morgan’s filter, Dijkstra’s paths, lattice cuts, bipartite graph (Figures A–D)

Data sources: `Shiny_Hunters_PeopleSoftZeroDay.json` (1,805 nodes, 2,186 links), `EntireGraph.json` (458 nodes), `jspWebShellIdentifiers.json` (60 hashes), `subset.csv` (20 domains), `VirusTotal report for LA.exe.pdf`, `Yomi Hunter.html.pdf`.

Dependencies: Python 3.12, `networkx`, `matplotlib`, `ipwhois`, `dnspython`, `python-whois` (see `requirements.txt`).